Banana Financial Intelligence Service

A production-grade RAG (Retrieval-Augmented Generation) system for grounded financial query answering with hallucination prevention, deployed on Kubernetes.

Prerequisites

Make sure the following are installed before you begin:

Tool	Version	Install
Docker	20+	https://docs.docker.com/get-docker
Minikube	1.32+	https://minikube.sigs.k8s.io/docs/start
kubectl	1.28+	https://kubernetes.io/docs/tasks/tools
Python	3.12	https://www.python.org/downloads

Step 1 — Get an Alpha Vantage API Key

The market data server requires a free API key from Alpha Vantage.

1. Go to https://www.alphavantage.co/support/#api-key
2. Fill in the short form — key is generated instantly
3. Export it in your terminal:

```bash
export ALPHA_VANTAGE_API_KEY=your_key_here
```

> Free tier: 25 requests/day, 5 requests/minute — enough for testing.

Step 2 — Get Social Sentiment API Keys (Optional)

The social server supports Twitter/X and Reddit for real sentiment analysis using VADER scoring. Without these keys the social source is skipped and the pipeline continues with market and SEC data only.

Twitter/X Bearer Token (primary)

1. Go to https://developer.twitter.com/en/portal/dashboard

2. Create a project and app (free Basic tier is sufficient)
3. Copy the **Bearer Token** from the Keys and Tokens tab
4. Export it:

```bash
export TWITTER_BEARER_TOKEN=your_bearer_token_here
```

Reddit API Credentials (fallback if Twitter unavailable)

1. Go to https://www.reddit.com/prefs/apps
2. Click **Create another app** → select **script**
3. Copy the **client ID** (under the app name) and **client secret**
4. Export them:

```bash
export REDDIT_CLIENT_ID=your_client_id_here
export REDDIT_CLIENT_SECRET=your_client_secret_here
```

> You only need one of Twitter or Reddit. Twitter is tried first; Reddit is the fallback.

Step 3 — Start Minikube

```bash
minikube start --cpus=4 --memory=6144 --disk-size=20g
```

Verify it is running:

```bash
minikube status
```

Step 4 — Create Kubernetes Secrets

Create the secrets before deploying so the pods can read them at startup.

Alpha Vantage secret (required)

```bash
```

```
kubectl create secret generic alpha-vantage-secret \
  --from-literal=ALPHA_VANTAGE_API_KEY=$ALPHA_VANTAGE_API_KEY
```

Social API secret (optional)

```bash
# Twitter + Reddit
kubectl create secret generic social-api-secret \
  --from-literal=TWITTER_BEARER_TOKEN=$TWITTER_BEARER_TOKEN \
  --from-literal=REDDIT_CLIENT_ID=$REDDIT_CLIENT_ID \
  --from-literal=REDDIT_CLIENT_SECRET=$REDDIT_CLIENT_SECRET

# Twitter only
kubectl create secret generic social-api-secret \
  --from-literal=TWITTER_BEARER_TOKEN=$TWITTER_BEARER_TOKEN

# Reddit only
kubectl create secret generic social-api-secret \
  --from-literal=REDDIT_CLIENT_ID=$REDDIT_CLIENT_ID \
  --from-literal=REDDIT_CLIENT_SECRET=$REDDIT_CLIENT_SECRET
```

> If you skip this step the social pod still starts — all secret refs are marked `optional: true` in the deployment YAML.

Step 5 — Deploy to Kubernetes

Run the deploy script from the project root:

```bash
chmod +x deploy.sh
./deploy.sh
```

The script will:

- Point Docker at Minikube's internal daemon
- Build all 4 images: `banana-api:v2`, `banana-market:v1`, `banana-sec:v1`, `banana-social-mcp:v1`
- Apply all manifests (ConfigMap, MCP services, API deployment)
- Wait for all pods to become ready

> **First run:** The API pod downloads FinBERT and TinyLlama on startup (~1–2 GB). Allow 3–5 minutes.

Step 6 — Verify All Pods Are Running

```bash
kubectl get pods
```

Expected output:

```
NAME                    READY   STATUS    RESTARTS
banana-api-xxx          1/1     Running   0
banana-market-xxx       1/1     Running   0
banana-sec-xxx          1/1     Running   0
banana-social-xxx       1/1     Running   0
```

If any pod is not `Running`, check its logs:

```bash
kubectl logs deployment/banana-api
kubectl logs deployment/banana-market
kubectl logs deployment/banana-sec
kubectl logs deployment/banana-social
```

Confirm the social server loaded correctly:

```bash
kubectl logs deployment/banana-social | grep -E "Tool registered|TWITTER|REDDIT|unavailable"
```

With keys set, requests will log:

```
Tool registered: fetch_social_sentiment
```

Without keys, the first request will log:

```
TWITTER_BEARER_TOKEN not set — skipping Twitter
REDDIT_CLIENT_ID/SECRET not set — skipping Reddit
```

This is fine — the pipeline continues with market and SEC data only.

Step 7 — Access the API

On macOS with the Docker driver, use `port-forward` for a stable local URL.

Open a dedicated terminal and keep it running:

```bash
kubectl port-forward service/banana-api-service 8080:80
```

The API is now available at `http://localhost:8080`.

Step 8 — Test the API

Quick smoke test

Open a second terminal and run:

```bash
curl -s -X POST http://localhost:8080/analyze \
  -H "Content-Type: application/json" \
  -d '{"query": "What is AAPL stock price?"}' | python3 -m json.tool
```

Expected response (with social keys configured):

```json
{
    "answer": "AAPL (2026-03-03) — Open: $263.32, High: $265.56, Close: $263.75 ...",
    "grounded": true,
    "hallucination_rate": 0.0,
    "faithfulness_score": 1.0,
    "block_reason": null,
    "tools_used": ["market", "sec", "social"],
    "sentiment": {
        "label": "neutral",
        "confidence": 0.921
    },
    "report": "\nFinancial Report:\n\nSentiment: neutral\nConfidence: 0.921\n"
}
```

Without social keys, `tools_used` will show `["market", "sec"]` only — this is expected.

Full integration test suite

```bash
python3 test_service.py
```

Runs 17 queries across 6 categories and prints a summary report.

Benchmark (baseline vs optimized)

```bash
python3 benchmark.py --url http://localhost:8080

# Quick run with 4 queries only
python3 benchmark.py --url http://localhost:8080 --max 4

# Save results to CSV
python3 benchmark.py --url http://localhost:8080 --out results.csv
```

Rebuilding After Code Changes

Rebuild the API

```bash
eval $(minikube docker-env)
docker build -t banana-api:v2 .
kubectl rollout restart deployment/banana-api
kubectl rollout status deployment/banana-api
```

Confirm the new code is running:

```bash
kubectl logs deployment/banana-api | grep -E "Analyst|Scribe|Reflection"
```

Expected:

```
[API] Initialising AnalystAgent (loading FinBERT)…
[API] AnalystAgent / ReflectionAgent / ScribeAgent ready ✓
```

Rebuild the social server

After any changes to `social_server.py` or `Dockerfile.social`:

```bash
eval $(minikube docker-env)
docker build -f mcp_servers/Dockerfile.social -t banana-social-mcp:v1 .
kubectl apply -f banana-social-deployment.yaml
kubectl rollout restart deployment/banana-social
kubectl rollout status deployment/banana-social
```

Update social API keys after deployment

```bash
# Delete and recreate the secret with new values
kubectl delete secret social-api-secret
kubectl create secret generic social-api-secret \
  --from-literal=TWITTER_BEARER_TOKEN=new_token \
  --from-literal=REDDIT_CLIENT_ID=new_id \
  --from-literal=REDDIT_CLIENT_SECRET=new_secret

# Restart the pod to pick up the new secret
kubectl rollout restart deployment/banana-social
```

Useful Commands

Check pod and service status

```bash
kubectl get pods
kubectl get services
```

Stream live logs

```bash
kubectl logs -f deployment/banana-api
kubectl logs -f deployment/banana-market
kubectl logs -f deployment/banana-sec
kubectl logs -f deployment/banana-social
```

Filter logs by component

```bash
kubectl logs deployment/banana-api | grep "\[Fetch\]"
kubectl logs deployment/banana-api | grep "\[Analyst\]"
kubectl logs deployment/banana-api | grep "BLOCKED"
kubectl logs deployment/banana-social | grep "Sentiment"
```

Verify secrets are set correctly

```bash
```

```
kubectl get secret alpha-vantage-secret -o jsonpath='{.data.ALPHA_VANTAGE_API_KEY}' | base64 --decode
kubectl get secret social-api-secret -o jsonpath='{.data.TWITTER_BEARER_TOKEN}' | base64 --decode
```

Restart a specific pod

```bash
kubectl rollout restart deployment/banana-api
kubectl rollout restart deployment/banana-social
```

Stop Minikube

```bash
minikube stop
```

Common Issues

Port 8080 already in use

```bash
lsof -i :8080
kill -9 <PID>
```

Pod stuck in Pending or CrashLoopBackOff

```bash
kubectl describe pod -l app=banana-api
kubectl logs deployment/banana-api --previous
```

New image not picked up after rebuild

Ensure you ran `eval $(minikube docker-env)` before `docker build`. Without this, Docker builds into your local daemon instead of Minikube's.

```bash
eval $(minikube docker-env)
docker images | grep banana    # check timestamps are recent
kubectl rollout restart deployment/banana-api
```

Market data returning empty results

Your Alpha Vantage key may not be set or the free tier daily limit (25 req/day) has been reached.

```bash
kubectl get secret alpha-vantage-secret -o jsonpath='{.data.ALPHA_VANTAGE_API_KEY}' | base64 --decode
```

Social sentiment not appearing in tools_used

Either the API keys are not configured, or the sentiment result was rejected by `validate_node`. Check:

```bash
kubectl logs deployment/banana-social --tail=20
```

If you see `TWITTER_BEARER_TOKEN not set` and `REDDIT_CLIENT_ID/SECRET not set`, the keys are missing from the secret. Recreate the secret and restart the pod (see *Update social API keys* above).

Configuration Reference

Variable	Source	Default	Description
`EXPERIMENT_MODE`	ConfigMap	`OPTIMIZED`	`BASELINE` (plain LLM) or `OPTIMIZED` (RAG + agents)
`CONFIDENCE_THRESHOLD`	main.py	`0.70`	FinBERT minimum confidence to generate a report
`SIMILARITY_THRESHOLD`	main.py	`0.55`	ChromaDB cosine similarity minimum for retrieved docs
`ALPHA_VANTAGE_API_KEY`	Secret	—	Required for live market data
`TWITTER_BEARER_TOKEN`	Secret	—	Optional — Twitter/X social sentiment (primary)
`REDDIT_CLIENT_ID`	Secret	—	Optional — Reddit social sentiment (fallback)
`REDDIT_CLIENT_SECRET`	Secret	—	Optional — Reddit social sentiment (fallback)
`REDDIT_USER_AGENT`	Deployment env	`BananaFinanceBot/1.0`	Reddit API user agent string

To switch to BASELINE mode, edit `banana-configmap.yaml`:

```yaml
```

```yaml
data:
  EXPERIMENT_MODE: "BASELINE"
```

Then apply:

```bash
kubectl apply -f banana-configmap.yaml
kubectl rollout restart deployment/banana-api
```

Local Docker Alternative (No Kubernetes)

If you prefer running without Minikube:

```bash
export ALPHA_VANTAGE_API_KEY=your_key_here
export TWITTER_BEARER_TOKEN=your_bearer_token_here   # optional
export REDDIT_CLIENT_ID=your_client_id               # optional
export REDDIT_CLIENT_SECRET=your_client_secret       # optional
docker-compose up --build
```

API will be available at `http://localhost:8000/analyze`.